Project Proposal: Text-to-Image Generation with CLIP and Stable Diffusion

Course	IE7615 Deep Learning for AI	**Group**	Group 1, Section 03
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		Project	Generative project

1. Objective

Build a text-to-image pipeline on a curated COCO image–caption subset: (a) a reproducible zero-shot Stable Diffusion v1.5 baseline, and (b) a test of whether lightweight LoRA fine-tuning improves prompt adherence on COCO-style scenes.

2. Dataset

- **Source:** MS-COCO 2017 captions (official annotations, CC BY 4.0), 5 human captions per image.
- **Curated subset:** 1,998 image–caption pairs (2,000 sampled, 2 duplicate captions removed), seeded random sample, split 1,598 train / 199 val / 201 test by image so no image crosses splits.
- **Cleaning:** ASCII normalisation, whitespace and punctuation fixes, drop captions under 4 or over 40 words (CLIP caps at 77 tokens), remove duplicate captions, require shorter image side ≥ 256 px; square centre-crop to 512×512.
- **Why COCO over Flickr30k:** broader object and scene variety and a direct official download; Flickr30k is people-centric with stricter access.

3. Model and approach

- **Architecture:** Stable Diffusion v1.5 latent diffusion. A frozen CLIP ViT-L/14 text encoder produces 77×768 token embeddings; a UNet denoises in the VAE latent space with cross-attention to the text.
- **Baseline (Phase 1):** zero-shot SD v1.5, DPM-Solver++ with 30 steps, guidance scale 7.5, five fixed prompt–seed pairs so every later model is compared on identical inputs.
- **Adaptation (Phase 2):** LoRA on UNet attention layers (rank 8–16) trained on the 1,598 train pairs in fp16. Ablations: single vs. all five captions per image, and LoRA rank.
- **Stack and compute:** PyTorch, Hugging Face Diffusers, Transformers (CLIP); fits a 16 GB GPU (Colab T4).

4. Evaluation

- CLIPScore on the 5 test prompts and the 201 held-out captions; retrieval recall on val as a data-quality check.
- FID/KID of generated vs. real test images (noted as noisy at n≈200), plus a small blind human preference comparison of baseline vs. LoRA outputs.
- **Deliverables:** this proposal, 5 sample prompt→image test runs, and the initialised GitHub repo (see Note below).

5. Team roles

Role	Owner	Responsibilities
Data lead	Aditi Kulkarni	Dataset curation, cleaning and preprocessing pipeline; split and dataset statistics.
Modeling lead	Aditya Sai Kumar Kusetti	Environment, SD v1.5 baseline, LoRA fine-tuning runs, hyperparameter ablations.
Evaluation lead	Sakshi Pawar	CLIPScore, retrieval and FID/KID metrics, fixed-seed prompt panel, human preference check.
Integration & docs lead	Hemanth Mareedu	GitHub repo, tests, README, results tracking, final report.

6. Note

The 5 sample prompt→image test runs are in the GitHub repo at **results/baseline_v0/grid.png** (all five together), with run_01.png to run_05.png as the individual images.

GitHub repo: https://github.com/hmareedu457-svg/text2image-coco-diffusion